CISCaRL

Complete Code and File Guide

Every code file and every project area, in plain language

Support-constrained interpretable subgroup discovery for survival outcomes

Dataset: ACTG175, arms 0 and 1, n = 1,054

Final model: C200-I · horizon: 730 days

Prepared: November 2026

This guide covers all 84 code files in the project, the 31 notebook steps, and every non-code area: data, results, figures, reports and configuration.

Coverage at a glance

Code files explained	84 of 84: ciscarl2 (74), ciscarl2/external (2), run_all.py (1), paper builders and content (7)
Notebook steps	31 of 31, each with code, plain explanation, result and decision
Result files	293 files: 121 Markdown records and 167 CSV tables
Paper figures	38 files: 19 PNG and 19 PDF versions
Report exhibits	42 images: 38 exact console screenshots and 4 notebook plots
Docs and config	README, requirements, .gitignore, LICENSE, data, notebook, report, paper

How to read each entry

Every code file is shown with its path, a status label, and three bullets: what it is, what it does, and what it produced. Historical gates are kept because they explain why a choice was made or rejected; they are not needed to run the final model.

Status labels

- **Final pipeline —** part of the model that produces the final rule or decision
- **Final evidence —** a gate or check that supports a final choice
- **Historical evidence —** an earlier experiment kept for the audit trail
- **Tested and rejected —** an alternative that was implemented, run and not adopted
- **Support —** helper code with no experimental output
- **External dependency —** the SCRE baseline; not used by the final model
- **Build tool —** code that rebuilds the notebook, report, figures or paper

Part A — Every code file

Core model

11 files in this group.

ciscarl2/paths.py — *Support*

- **What it is:** A small locator for the data file.
- **What it does:** Checks the ACTG175_PATH variable, then data/ACTG175.csv, then the root copy, and returns one path.
- **What it produced:** Resolves to data/ACTG175.csv. The root and data-folder copies have the same SHA-256 checksum.

ciscarl2/__init__.py — *Support*

- **What it is:** The package marker for ciscarl2.
- **What it does:** Tells Python that the folder is a package and records the project name.
- **What it produced:** No output. It exists so every other module can be imported as ciscarl2.*.

ciscarl2/audit.py — *Final evidence*

- **What it is:** A small self-test suite for the rule code and the estimator.
- **What it does:** Checks that rule conditions match patient masks, that tree leaves do not overlap, that the influence-function contrast equals the direct formula, and that purity, capture and Jaccard match hand counts.
- **What it produced:** All checks pass. AIPW bias is +0.0031 (SD 0.0130) and the arm means are within 0.002.

ciscarl2/audit_commit.py — *Final evidence*

- **What it is:** A check that the commitment decision is not scored on the same patients used to make it.
- **What it does:** Compares deciding on all test patients with deciding on one half and scoring on the other.
- **What it produced:** Benefit per treated patient is 0.1975 full-sample and 0.1978 honest-split; efficiency is 0.987 vs 0.989.

ciscarl2/dgp.py — *Final evidence*

- **What it is:** The simulation data generator.
- **What it does:** Resamples the real ACTG175 covariate matrix, randomises treatment 50/50, generates survival and censoring times, and adds a known effect in one or two regions.
- **What it produced:** Prevalence 25.5%, event rate 20.0%, censoring 10.1%, true ATE +0.050, true regional effect +0.195 and 0 outside.

ciscarl2/diagnostics.py — *Final evidence*

- **What it is:** The nuisance diagnostic report for the real data.
- **What it does:** Runs proportional-hazards tests, tie handling checks, survival calibration, and weight-cap sensitivity, then writes diagnostics.md and three CSV files.

- **What it produced:** Global PH p = 0.235 / 0.368; cd40 p = 0.021 / 0.023; calibration ECE 0.032; cap 1.2 binds 9.9% of weights and caps of 1.5 or more never bind.

ciscarl2/estimators.py — *Final pipeline*

- **What it is:** The production estimator.
- **What it does:** Fits a survival model for each arm and a censoring model, cross-fits over five folds, builds the AIPW pseudo-outcome, and caps extreme weights.
- **What it produced:** Used by every gate. Audit bias +0.0031; cap is min(99th percentile, 20); effective sample size on discovery is 372 of 527.

ciscarl2/rules.py — *Final pipeline*

- **What it is:** The rule representation and cleanup code.
- **What it does:** Turns each tree leaf into a readable if-then rule, removes duplicate conditions, prunes broad rules, and assigns patients to the most specific match.
- **What it produced:** Audit shows conditions equal their patient masks exactly, and leaves partition the data.

ciscarl2/nested_tune.py — *Final evidence*

- **What it is:** Hyperparameter tuning with honest nested refitting.
- **What it does:** Tunes depth, leaf size and split criterion by cross-validation while refitting the nuisance models inside every fold.
- **What it produced:** Best nested MAE: depth 4 / leaf 60 / squared 0.8633; frozen depth 3 is 0.8638; absolute error is 0.8711. The earlier 0.824 was leakage and was withdrawn.

ciscarl2/policy_eval.py — *Final evidence*

- **What it is:** Policy value and net-benefit comparison.
- **What it does:** Compares treat-none, C200, SCRE and treat-all, including a cost per treated patient and influence-function coverage.
- **What it produced:** Policy value: none 0.7659; C200 0.7870 single / 0.7855 two; SCRE 0.7926 / 0.8044; treat-all 0.8145 / 0.8345. C200 leads net benefit from a cost of 0.05–0.10; IF coverage 92.5%.

ciscarl2/scre_vs_ciscarl.py — *Final evidence*

- **What it is:** The real-data rule table and SCRE comparison.
- **What it does:** Lists every CART leaf with support, validation and assessment effects, and compares C200 with SCRE on the same patients.
- **What it produced:** Selected rule cd40≤407.5 & wtkg>71.0 & homo≤0.5; support 61; validation +0.116 (LCB −0.250); assessment +0.401 on 38 patients; decision INSUFFICIENT.

Gates 1–4

12 files in this group.

ciscarl2/gate1.py — *Historical evidence*

- **What it is:** The first estimator validation.

- **What it does:** Tests AIPW under four censoring mechanisms and three censoring levels, with 200 replications per cell and a known effect.
- **What it produced:** Bias is small at 10–25% censoring but −0.0302 at 50% covariate censoring; event-dependent censoring bias grows from −0.0152 to −0.0806.

ciscarl2/gate1_verify.py — *Historical evidence*

- **What it is:** A check that the weight cap still works under the full cross-fitting pipeline.
- **What it does:** Compares the single cross-fit used in Gate 1b with the outer five-fold production pipeline.
- **What it produced:** Confirms the cap behaves as intended once the production folds are used.

ciscarl2/gate1b.py — *Historical evidence*

- **What it is:** The weight-stabilisation experiment.
- **What it does:** Compares no cap, fixed caps at 50/20/10, the 99th and 95th percentiles, and an oracle cap.
- **What it produced:** Uncapped maximum weight 1,000 and RMSE 0.2942; a cap at 20 gives RMSE 0.1172; the 99th percentile gives 0.0983. The project adopted min(99th percentile, 20).

ciscarl2/gate1c.py — *Historical evidence*

- **What it is:** A check of the Cox comparator under non-proportional hazards.
- **What it does:** Runs a crossing-hazards design and compares a pooled Cox model with the separate-arm model.
- **What it produced:** The pooled Cox proxy is biased by 16–45% under crossing hazards, so every comparison uses the separate-arm production model.

ciscarl2/gate2.py — *Historical evidence*

- **What it is:** Rule-level inference, abstention and stability.
- **What it does:** Measures interval coverage, the fraction of runs with no rule, and TREAT sign accuracy.
- **What it produced:** Coverage 88.3% over 103 rules in the first run; 15–30% of replications returned no rule. The coverage result was later rechecked with more replications in Gate 2b.

ciscarl2/gate2_funnel.py — *Historical evidence*

- **What it is:** A discovery funnel for the no-rule cases.
- **What it does:** Records where candidate rules are lost: raw leaves, after pruning, after support, and after stability.
- **What it produced:** Identifies the stage that removes candidates. The result is saved in results/gate2_funnel*.csv.

ciscarl2/gate2b.py — *Historical evidence*

- **What it is:** A higher-precision check of subgroup intervals.
- **What it does:** Repeats the interval study with 100 replications and compares the plain influence-function interval with an EIF-corrected version.
- **What it produced:** Coverage is 93.1% over 333 rules; the nuisance correction changes variance by only 1–2%. The earlier 88.3% was mostly small-sample noise.

ciscarl2/gate2c.py — *Historical evidence*

- **What it is:** A sweep of the stability-selection threshold.
- **What it does:** Runs nested stability thresholds and measures discovery quality versus calibration.
- **What it produced:** The project chose the 0.50 threshold; the sweep is recorded in results/gate2c_summary.csv.

ciscarl2/gate2d.py — *Historical evidence*

- **What it is:** An application-matched sample-size test.
- **What it does:** Repeats the pipeline at n = 589, which is the real discovery size, with the same effect scale and censoring as ACTG175.
- **What it produced:** Shows how little power the real sample gives; this is the reason accuracy claims are made in simulation, not on ACTG175.

ciscarl2/gate2e.py — *Historical evidence*

- **What it is:** A diagnosis of the decision layer at the real sample size.
- **What it does:** Measures false commitment among rules whose true effect is near zero and sweeps the commitment threshold.
- **What it produced:** Near-zero rules are committed 6.5–8.3% of the time against a nominal 5%; no threshold gives 90% precision and at least 50% commitment.

ciscarl2/gate3.py — *Historical evidence*

- **What it is:** The delta sensitivity analysis under a known missing-not-at-random mechanism.
- **What it does:** Checks the delta = 0 identity and tests whether tilting the outcome model recovers the truth.
- **What it produced:** Delta = 0 reproduces the baseline to 1.7e-16. The adjustment slope is about 0.002 per unit delta, far too small to correct a bias of 0.01–0.02.

ciscarl2/gate4.py — *Historical evidence*

- **What it is:** The first comparison with the real SCRE implementation.
- **What it does:** Matches data, split, horizon and evaluation patients, and fits both methods on the same seeds.
- **What it produced:** Fixes a degenerate SCRE alpha choice and sets the comparison protocol used later. The final head-to-head is reported in Gate 28.

Gates 5–15

11 files in this group.

ciscarl2/gate5.py — *Historical evidence*

- **What it is:** Subgroup-level missing-not-at-random sensitivity.
- **What it does:** Tilts the censoring weights over $\delta \in [-2, 2]$ and records where a TREAT verdict changes.
- **What it produced:** All three TREAT verdicts survive the whole range, moving by about ±0.02. The production cap of 1.31 absorbs the tilt, which is reported as a limitation.

ciscarl2/gate6.py — *Historical evidence*

- **What it is:** A test of whether the problem is the pseudo-outcome or the rule extraction.
- **What it does:** Runs the same extraction and selection on a true-effect target and on the estimated target.
- **What it produced:** With a clean target the rule machinery recovers the region, so extraction is not the main bottleneck; the target is.

ciscarl2/gate7.py — *Historical evidence*

- **What it is:** A minimum-support sweep at n = 2000.
- **What it does:** Varies the support floor and compares the rule model with the constant baseline.
- **What it produced:** The sweep is stored in results/gate7_summary.csv and informed the 200-patient floor.

ciscarl2/gate8.py — *Tested and rejected*

- **What it is:** A test of the RMST estimand.
- **What it does:** Rebuilds every step for restricted mean survival time and compares it with the constant.
- **What it produced:** RMST is worse than its own constant by 3.495 days (+10.8% MAE, CI above zero). It was not adopted.

ciscarl2/gate9.py — *Tested and rejected*

- **What it is:** A test of a smoother plug-in discovery target.
- **What it does:** Compares the AIPW target with Cox and RSF plug-in targets while keeping the final evaluation on held-out AIPW.
- **What it produced:** The plug-in target is 10–14 times less noisy and lifts Jaccard from 0.107 to about 0.61, but final MAE is worse than the constant (+0.0047 / +0.0094). It was not adopted.

ciscarl2/gate10.py — *Historical evidence*

- **What it is:** Conservative rule extraction on the cross-fitted plug-in target.
- **What it does:** Builds candidate rules from a tree on the plug-in target and measures their support and stability.
- **What it produced:** The result is saved in results/gate10_*.csv and was an early step before the AIPW target was adopted.

ciscarl2/gate11.py — *Historical evidence*

- **What it is:** The first sample-size result for the depth-2 rule model.
- **What it does:** Compares the rule model with the constant at 4,000, 8,000 and 12,000 patients, 40 seeds per cell.
- **What it produced:** Gain is +9.2% at 4,000, +19.6% at 8,000 and +20.6% at 12,000. The model is adopted at 8,000 and 12,000 but not at 4,000.

ciscarl2/gate12.py — *Historical evidence*

- **What it is:** The confirmation of the depth-3 rule model on fresh seeds.
- **What it does:** Repeats the depth-3 comparison at 8,000 and 12,000 patients with new seeds.

- **What it produced:** Gain is +26.0% at 8,000 and +33.7% at 12,000, with purity 0.87–0.95. Depth 3 is adopted.

ciscarl2/gate13.py — *Historical evidence*

- **What it is:** A delta-tilt sensitivity on the adopted rule set.
- **What it does:** Discovers the rule once at delta = 0, then re-estimates it under several censoring tilts.
- **What it produced:** The estimates move only slightly across the tilt range; the details are in results/gate13_tilt.csv.

ciscarl2/gate14.py — *Historical evidence*

- **What it is:** A test of the rule model under a misspecified Cox outcome model.
- **What it does:** Runs the pipeline with a correct, mild and severe outcome-model violation.
- **What it produced:** Gain is +30.1% at κ=0 and +23.8% at κ=1, but −4.1% at κ=3. The method is not robust to severe outcome-model misspecification.

ciscarl2/gate15.py — *Tested and rejected*

- **What it is:** A test of blending the plug-in target with AIPW.
- **What it does:** Standardises both targets and mixes them at several weights under a severe violation.
- **What it produced:** The blend fails (MAE +0.0022, 48% of replications better); AIPW alone is better and more robust.

Gates 16–30

13 files in this group.

ciscarl2/gate16.py — *Historical evidence*

- **What it is:** The confirmation of the AIPW target.
- **What it does:** Compares AIPW with the plug-in target at 8,000 and 12,000 patients, with correct and wrong outcome models.
- **What it produced:** AIPW gains 37.8%/48.0% at κ=0 and 32.7%/53.4% at κ=3; the plug-in falls to −4.0%/+6.5% at κ=3. AIPW is adopted.

ciscarl2/gate17.py — *Historical evidence*

- **What it is:** The lower sample-size boundary of the final design.
- **What it does:** Compares the AIPW rule model, its constant and an oracle at 2,000, 4,000 and 6,000 patients.
- **What it produced:** At 2,000 it is worse (−22.8%); at 4,000 it is +8.1% and at 6,000 +23.0%. The gap to the oracle is 40–61 points.

ciscarl2/gate18.py — *Historical evidence*

- **What it is:** A joint outcome-and-censoring misspecification test.
- **What it does:** Crosses a correct or wrong outcome model with a conditional or marginal censoring model.

- **What it produced:** Prediction still gains about 30%, but subgroup purity falls from 0.823 to 0.659 in the worst cell: average prediction survives, subgroup reliability does not.

ciscarl2/gate19.py — *Final evidence*

- **What it is:** The commitment-versus-shrinkage experiment.
- **What it does:** Compares no commitment, hard commitment at a positive lower bound, and shrinkage only.
- **What it produced:** Hard commitment removes null-benefit treatment (10–12% to 0%), raises benefit per treated patient to +0.176/+0.183, and abstains in 30–45% of runs.

ciscarl2/gate20.py — *Final evidence*

- **What it is:** The honest commitment test with disjoint roles.
- **What it does:** Uses 4,000 discovery, 2,000 validation and 2,000 assessment patients, and repeats with only 1,000 validation patients.
- **What it produced:** Full validation gives 50% commitment, 0% null treatment and MAE −0.0131 vs constant; half validation gives MAE +0.0083. Validation size matters.

ciscarl2/gate21.py — *Final evidence*

- **What it is:** The selection-rule comparison.
- **What it does:** Compares max mean, lower confidence bound, support at least 200 and stability-weighted selection.
- **What it produced:** Support 200 captures 0.548 vs 0.356, raises purity 0.69→0.81, Jaccard 0.33→0.51, and removes null-benefit leaves (5%→0%). It is adopted.

ciscarl2/gate22.py — *Historical evidence*

- **What it is:** A valid stability measurement.
- **What it does:** Refits each configuration with its own selection procedure on 20 disjoint discovery half-samples.
- **What it produced:** C200 has Jaccard 0.318/0.334 and capture 0.623; every configuration produces many different rules across splits.

ciscarl2/gate23.py — *Final evidence*

- **What it is:** The two-region confirmation of the support floor.
- **What it does:** Compares max mean, support 100/200/400 and a stability rule on a two-region effect.
- **What it produced:** C200 captures 0.364 vs 0.148 (+0.216), recovers Jaccard 0.295 vs 0.127, keeps purity 0.82 vs 0.77, and commits 70% vs 45%.

ciscarl2/gate24.py — *Historical evidence*

- **What it is:** A joint nuisance stress test on the two-region design.
- **What it does:** Adds control-arm non-proportional hazards and censoring misspecification to the two-region effect.
- **What it produced:** Prediction holds, but subgroup recovery degrades; the numbers are in results/gate24_summary.csv.

ciscarl2/gate25.py — *Tested and rejected*

- **What it is:** The two-leaf rule-list experiment.
- **What it does:** Compares the single C200 leaf with a two-leaf list from the same tree.
- **What it produced:** The list captures more (+0.169) but purity falls by 0.166 and per-treated benefit by 0.009. It was rejected; one leaf is kept.

ciscarl2/gate26.py — *Tested and rejected*

- **What it is:** The flexible DR-learner with rule distillation.
- **What it does:** Learns a more complex effect surface and distils a simple rule from it.
- **What it produced:** It is purer (+0.068/+0.058) but captures less (−0.094/−0.073) and has slightly worse MAE. It was not adopted.

ciscarl2/gate28.py — *Final evidence*

- **What it is:** The head-to-head with SCRE on identical splits.
- **What it does:** Fits both methods on the same discovery, validation and assessment patients and the same horizon.
- **What it produced:** C200 MAE is worse by 0.0043 (p=0.046) in one region and 0.0031 (p=0.112) in two; purity is 0.86/0.87 vs 0.43/0.58; capture is 0.59/0.33 vs 0.95/0.80.

ciscarl2/gate30.py — *Final evidence*

- **What it is:** The C200 knob sweep and confirmation.
- **What it does:** Tests shrinkage, two commitment thresholds, two leaf sizes and depth 4 on 40 tuning and 40 fresh seeds.
- **What it produced:** Shrinkage lowers MAE by 0.0054/0.0025; depth 4 is worse (+0.0092/+0.0073); leaf size and threshold do almost nothing. Shrinkage becomes C200-S.

Gates 31–44

14 files in this group.

ciscarl2/gate31.py — *Final evidence*

- **What it is:** The outcome-nuisance comparison.
- **What it does:** Compares per-arm Cox, random survival forest, gradient boosting and equal-weight stacking.
- **What it produced:** Cox has the best Brier (0.0028/0.0033). Stacking lowers MAE by 0.0043/0.0075 but loses 0.030 purity and costs 136–260 s. Cox is retained.

ciscarl2/gate32.py — *Final evidence*

- **What it is:** The rule-learner comparison.
- **What it does:** Compares CART, pruned CART, PRIM, a classification tree and boosting distillation.
- **What it produced:** Pruned CART purity falls to 0.741/0.637, PRIM to 0.328/0.473, classification to 0.671/0.701, boosting to 0.883/0.761. CART (0.889/0.811) is retained.

ciscarl2/gate33.py — *Final evidence*

- **What it is:** The calibration and interval comparison.

- **What it does:** Compares no calibration, shrinkage and isotonic; and influence-function, bootstrap, selection bootstrap and conformal intervals.
- **What it produced:** Isotonic lowers MAE by 0.0053/0.0065 on fresh seeds and is adopted for prediction (C200-I). The selection bootstrap fails (85%) and conformal is 2.33× wider; IF is retained.

ciscarl2/gate34.py — *Final evidence*

- **What it is:** The fold-count comparison.
- **What it does:** Compares 3, 5 and 10 cross-fitting folds with everything else fixed.
- **What it produced:** Ten folds gains 0.0005 MAE at 2.1–2.3× cost; three folds is worse by 0.0010–0.0011. Five folds are retained.

ciscarl2/gate35.py — *Final evidence*

- **What it is:** The fairness, SHAP and transport analysis.
- **What it does:** Describes rule effects by race and gender, measures calibration and SHAP, and transports a white-group rule to the non-white group.
- **What it produced:** Every group has <30 rule patients: white +0.444 (24), non-white +0.328 (14), male +0.514 (25), female +0.184 (13); female ECE 0.058; transport support 74, effect +0.223.

ciscarl2/gate36.py — *Tested and rejected*

- **What it is:** The fixed-horizon IPCW-TMLE comparison.
- **What it does:** Implements TMLE with the same folds and nuisances and compares it with AIPW.
- **What it produced:** TMLE MAE is 0.1408/0.1374 vs AIPW 0.0516/0.0578, Brier is 0.0177/0.0172 vs 0.0026/0.0031, and bias is larger. AIPW is retained.

ciscarl2/gate37.py — *Final evidence*

- **What it is:** The five-horizon screen.
- **What it does:** Compares 365, 548, 730, 912 and 1,095 days on events, at-risk share, support and precision.
- **What it produced:** No horizon meets every criterion. 730 has 99 events and 71% at risk; 365/548 have too few events; 1,095 has 51% censoring. 730 is kept as a compromise.

ciscarl2/gate38.py — *Final evidence*

- **What it is:** The feature-selection comparison.
- **What it does:** Compares 12 covariates, 16 candidates, lasso and RFE, all fitted on discovery only.
- **What it produced:** The 12- and 16-feature rules are identical; lasso keeps all 12; RFE keeps 11 and selects cd80>805 & race>0.5 with support 79. The 12 covariates are kept.

ciscarl2/gate39.py — *Tested and rejected*

- **What it is:** The learned-stacking comparison.
- **What it does:** Fits positive weights on Cox, RSF and GBM out-of-fold predictions and compares them with Cox and equal weights.

- **What it produced:** Learned weights are about 0.77 Cox / 0.14 RSF / 0.09 GBM and the result is indistinguishable from Cox (+0.0007/+0.0001). Learned stacking is rejected.

ciscarl2/gate40.py — *Final evidence*

- **What it is:** The SCRE alpha-path sensitivity.
- **What it does:** Fits SCRE once per replication and evaluates every alpha on the path on the same patients.
- **What it produced:** The chosen alpha is within 0.0010–0.0017 MAE of the best; the direction (SCRE more accurate, C200 purer) is unchanged.

ciscarl2/gate41.py — *Final evidence*

- **What it is:** The selection-aware interval comparison.
- **What it does:** Compares the influence-function interval with Bonferroni and sample-splitting.
- **What it produced:** IF covers 95–100% at width 0.297–0.298; Bonferroni covers 100% at 1.38–1.39× width; sample-splitting covers 100% at the same width but uses a different rule. IF is retained.

ciscarl2/gate42.py — *Final evidence*

- **What it is:** The dense horizon grid.
- **What it does:** Searches every 30 days from 365 to 1,085 and ranks by validation lower bound with event and support guards.
- **What it produced:** The single-split optimum is 965 days (LCB 0.000, effect +0.427, support 61, assessment +0.200).

ciscarl2/gate43.py — *Final evidence*

- **What it is:** The discovery-only horizon selection.
- **What it does:** Uses five-fold inner cross-validation inside discovery, never validation or assessment.
- **What it produced:** The optimum is 845 days (inner LCB +0.014 vs −0.118 for 730); validation +0.114, assessment +0.498.

ciscarl2/gate44.py — *Final evidence*

- **What it is:** The horizon stability test.
- **What it does:** Repeats 730, 845 and 965 on 20 different random splits and records validation and assessment.
- **What it produced:** 730: mean LCB −0.142, SD 0.114; 845: −0.167, SD 0.185; 965: −0.206, SD 0.191. 730 is retained on stability.

Gates 45–48

4 files in this group.

ciscarl2/gate45.py — *Tested and rejected*

- **What it is:** The broad CART tuning experiment.
- **What it does:** Runs 540 configurations over depth, leaf size, split criterion and pruning, with the nuisance models refitted inside every outer fold.

- **What it produced:** Best nested MAE is 0.85982 for depth 5 / leaf 30 / squared error, against 0.86379 for the frozen depth 3 / leaf 60 (rank 54 of 540). It was not adopted on prediction error alone.

ciscarl2/gate46.py — *Tested and rejected*

- **What it is:** The ensemble-versus-CART comparison.
- **What it does:** Distils random forest, extra trees, gradient boosting, histogram boosting and a ridge stack into a depth-3 rule, then applies the same support and commitment protocol.
- **What it produced:** No ensemble gains capture with a 95% interval above zero; costs are 4.8–65.9 s against 0.07 s for CART. CART is retained.

ciscarl2/gate47.py — *Tested and rejected*

- **What it is:** CART tuning on simulated subgroup performance.
- **What it does:** Evaluates 48 CART configurations on the same 20 splits × 2 DGPs, selecting on validation LCB with purity, capture and interpretability guardrails.
- **What it produced:** Depth 2 / absolute error is best in simulation (LCB +0.0605/+0.0731), but on ACTG175 it selects 122 patients, LCB −0.210 and abstains. The frozen tree remains.

ciscarl2/gate48.py — *Final evidence*

- **What it is:** The SHAP and LIME explainability audit.
- **What it does:** Runs exact TreeSHAP on the final CART, LIME on 40 rule-in/rule-out patients with five seeds, and compares top features, stability and group-wise SHAP.
- **What it produced:** SHAP: cd40 0.128, homo 0.093, cd80 0.057, wtkg 0.035; all other features zero. SHAP–LIME overlap 0.767/0.700; LIME stability 0.703/0.712. No model change.

Diagnostics and variants

9 files in this group.

ciscarl2/gate4c.py — *Historical evidence*

- **What it is:** The crossing-hazards comparison with real SCRE.
- **What it does:** Runs a crossing-hazards design against SCRE with matched splits and reconciles the earlier marginal and paired calculations.
- **What it produced:** The sign convention and paired subset are fixed; the comparison is internally consistent.

ciscarl2/gate4d.py — *Historical evidence*

- **What it is:** The piecewise-constant DGP comparison with SCRE.
- **What it does:** Compares CISCaRL and SCRE on a rule-shaped effect at 589 and 2,000 patients.
- **What it produced:** The rule-shaped advantage is not supported: sensitivity is 0.11 at 589 and 0.28 at 2,000, and the false-positive rate is 0.321 at 2,000.

ciscarl2/gate4d_cart.py — *Historical evidence*

- **What it is:** A three-way recoverability test.
- **What it does:** Compares a tree fitted to the true effect, a tree fitted to the pseudo-outcome, and the full search.

- **What it produced:** Shows whether the true rule is in the candidate pool and whether the selection step can find it.

ciscarl2/gate4d_diagnose.py — *Historical evidence*

- **What it is:** A failure-location diagnostic.
- **What it does:** Separates failure in the estimate, the interval and the decision for each discovered rule.
- **What it produced:** Records conditions, test support, in-region fraction and effect error, so the failure can be attributed rather than guessed.

ciscarl2/gate4d_discovery.py — *Historical evidence*

- **What it is:** A discovery-funnel diagnostic.
- **What it does:** Records where the true region is lost: representable, generated, or filtered by support or stability.
- **What it produced:** The output is in results/gate4d_discovery*.csv and explains the zero-rule cases.

ciscarl2/gate4d_stability_check.py — *Historical evidence*

- **What it is:** A stability-selection diagnostic.
- **What it does:** Checks whether a good rule was found but then rejected by the stability filter.
- **What it produced:** Separates discovery failure from stability-filter failure.

ciscarl2/gate4e.py — *Historical evidence*

- **What it is:** A variance-normalised splitting criterion.
- **What it does:** Tests whether normalising the split criterion makes the true region more reproducible.
- **What it produced:** The result is recorded in results/gate4e_summary.csv and informs the later n-sweep.

ciscarl2/gate4f.py — *Historical evidence*

- **What it is:** A larger-n recoverability test.
- **What it does:** Repeats the half-sample stability test at 589, 2,000, 4,000 and 8,000 patients.
- **What it produced:** At 8,000 the region is recovered (Jaccard 0.88, precision 0.94, recall 0.93) but stability is only 46–47.5% against a 50% target.

ciscarl2/gate4g.py — *Historical evidence*

- **What it is:** The final recoverability validation.
- **What it does:** Compares the production random forest, an oracle CART and a CART on the pseudo-outcome at 8,000 and 12,000 patients.
- **What it produced:** Production RF has Jaccard 0.86 and stability 37.6%, below the oracle's 61.4%; the gap is discovery, not representability.

External and build

10 files in this group.

ciscarl2/external/scre.py — *External dependency*

- **What it is:** The SCRE comparator implementation.
- **What it does:** Fits a survival causal rule ensemble with a Coxnet path and rule-interaction terms; used by Gates 4, 28 and 40.
- **What it produced:** It is the baseline only. The final CISCaRL model does not depend on it.

ciscarl2/external/__init__.py — *Support*

- **What it is:** The package marker for ciscarl2.external.
- **What it does:** Allows SCRE to be imported as a normal subpackage.
- **What it produced:** No output.

run_all.py — *Build tool*

- **What it is:** The one-command reproducibility runner.
- **What it does:** Runs the notebook, Gates 23–44, the figure builder and optionally the paper; backs up results before a full run and uses a sandbox for --quick.
- **What it produced:** The quick sandbox ran all 16 gate groups in 12.5 minutes; Gate 37 re-ran byte-for-byte.

paper/make_paper_figures.py — *Build tool*

- **What it is:** The paper figure generator.
- **What it does:** Reads the saved result CSVs and draws all paper figures.
- **What it produced:** Writes 15 PNG/PDF figure pairs into paper/figs.

paper/build_ieee_detailed.py — *Build tool*

- **What it is:** The detailed IEEE paper builder.
- **What it does:** Builds the two-column paper from the result numbers and figures.
- **What it produced:** Writes an 8-page paper with 15 figures and 10 tables.

paper/build_model_creation_detailed.py — *Build tool*

- **What it is:** The detailed 31-step report builder.
- **What it does:** Builds the 42-page report, inserting the exact output images and saved figures.
- **What it produced:** Writes the detailed model-creation report; 31 steps and 52 exhibits.

paper/report_content.py — *Build tool*

- **What it is:** The text content of the 31-step report.
- **What it does:** Stores each step's title, code, explanation, result, meaning and decision, plus the figure mapping.
- **What it produced:** No standalone output; imported by the detailed report builder.

paper/make_step_exhibits.py — *Build tool*

- **What it is:** The exact-output image generator.
- **What it does:** Reads the executed notebook, captures each step's printed output, and renders it as a console-style PNG.
- **What it produced:** Writes 38 console images and 4 notebook plots into paper/exhibits.

paper/build_code_guide.py — *Build tool*

- **What it is:** The builder for this complete code and file guide.
- **What it does:** Reads the entry list, writes the grouped explanations, the 80-file coverage checklist and the glossary, then saves the Word document.
- **What it produced:** Writes this guide. Its checklist is the proof that every code file is covered.

paper/code_guide_content.py — *Support*

- **What it is:** The content store for this guide.
- **What it does:** Holds one entry per code file: path, status, what it is, what it does, and what it produced.
- **What it produced:** No standalone output; imported by the guide builder.

Part B — Every non-code project area

Data and configuration

- **ACTG175.csv:** the raw trial file at the project root: 1,054 patients in arms 0/1, 27 columns; SHA-256 debab27b…c7141a.
- **data/ACTG175.csv:** the same file at the path used by the code; both copies have the same checksum.
- **README.md:** the project guide: layout, install, run commands, seeds and limits.
- **requirements.txt:** pinned Python 3.11 packages: numpy 2.4.6, pandas 2.2.3, scikit-learn 1.9.1, scikit-survival 0.28.0, lifelines 0.30.3, and the rest.
- **.gitignore:** keeps cache, virtual-environment and temporary files out of Git.
- **LICENSE:** the MIT licence for the code.

Notebook

- **notebooks/CISCaRL_31_steps.ipynb:** the 31-step executable record: 136 cells, 52 code cells, 0 error cells; runs in 1–2 minutes in evidence mode.
- **notebooks/CISCaRL_31_steps.html:** the same notebook exported for reading without Jupyter.

Results

- **results/CANONICAL_RESULTS.md:** the single source of truth for every reported number.
- **results/CLAIMS_v1.17.md:** the final list of defensible and withdrawn claims.
- **results/EVIDENCE_EXTENSION.md:** the Gates 31–44 summary.
- ***_PREREGISTRATION.md:** the question, criteria and decision rule written before each gate ran.
- ***_verdict.md:** the measured finding and decision for each gate.
- ***_summary.csv:** the average result per configuration; *replications.csv holds the raw runs.
- **diagnostics_*.csv:** proportional hazards, calibration, ties and weight-cap diagnostics.
- **sweep1000_*.csv:** 1,000-replication assessment sweeps; nested_tuning.csv and split_study_*.csv hold the tuning and split-ratio studies.
- **results/historical/:** three pre-canonical ACTG175 records, kept only because FINAL_AUDIT.md cites them; their numbers are superseded.

Paper and figures

- **paper/CISCaRL_Engineering_Report_v2.docx/.pdf:** the 40-page engineering report.
- **paper/CISCaRL_IEEE_conference_detailed.docx/.pdf:** the 8-page IEEE-style paper: 15 figures, 10 tables.
- **paper/CISCaRL_Model_Creation_31_Steps_Detailed.docx/.pdf:** the 42-page step-by-step report with 52 exact exhibits.
- **paper/figs/:** 38 files: 19 PNG and 19 PDF figures generated from the result CSVs.
- **paper/exhibits/:** 42 images used by the 31-step report: 38 console screenshots and 4 plots.
- **paper/*.py:** the builders and content files for the figures, the IEEE paper and the 31-step report.

Part C — Coverage checklist

Every code file appears below. The letter is its status: F = final pipeline, E = final evidence, H = historical, R = tested and rejected, S = support, X = external, B = build.

File	S	File	S
ciscarl2/paths.py	S	ciscarl2/gate24.py	H
ciscarl2/__init__.py	S	ciscarl2/gate25.py	R
ciscarl2/audit.py	E	ciscarl2/gate26.py	R
ciscarl2/audit_commit.py	E	ciscarl2/gate28.py	E
ciscarl2/dgp.py	E	ciscarl2/gate30.py	E
ciscarl2/diagnostics.py	E	ciscarl2/gate31.py	E
ciscarl2/estimators.py	F	ciscarl2/gate32.py	E
ciscarl2/rules.py	F	ciscarl2/gate33.py	E
ciscarl2/nested_tune.py	E	ciscarl2/gate34.py	E
ciscarl2/policy_eval.py	E	ciscarl2/gate35.py	E
ciscarl2/scre_vs_ciscarl.py	E	ciscarl2/gate36.py	R
ciscarl2/gate1.py	H	ciscarl2/gate37.py	E
ciscarl2/gate1_verify.py	H	ciscarl2/gate38.py	E
ciscarl2/gate1b.py	H	ciscarl2/gate39.py	R
ciscarl2/gate1c.py	H	ciscarl2/gate40.py	E
ciscarl2/gate2.py	H	ciscarl2/gate41.py	E
ciscarl2/gate2_funnel.py	H	ciscarl2/gate42.py	E
ciscarl2/gate2b.py	H	ciscarl2/gate43.py	E
ciscarl2/gate2c.py	H	ciscarl2/gate44.py	E
ciscarl2/gate2d.py	H	ciscarl2/gate4c.py	H
ciscarl2/gate2e.py	H	ciscarl2/gate4d.py	H
ciscarl2/gate3.py	H	ciscarl2/gate4d_cart.py	H
ciscarl2/gate4.py	H	ciscarl2/gate4d_diagnose.py	H
ciscarl2/gate5.py	H	ciscarl2/gate4d_discovery.py	H
ciscarl2/gate6.py	H	ciscarl2/gate4d_stability_check.py	H
ciscarl2/gate7.py	H	ciscarl2/gate4e.py	H
ciscarl2/gate8.py	R	ciscarl2/gate4f.py	H
ciscarl2/gate9.py	R	ciscarl2/gate4g.py	H
ciscarl2/gate10.py	H	ciscarl2/gate45.py	R
ciscarl2/gate11.py	H	ciscarl2/gate46.py	R
ciscarl2/gate12.py	H	ciscarl2/gate47.py	R
ciscarl2/gate13.py	H	ciscarl2/gate48.py	E
ciscarl2/gate14.py	H	ciscarl2/external/scre.py	X
ciscarl2/gate15.py	R	ciscarl2/external/__init__.py	S
ciscarl2/gate16.py	H	run_all.py	B
ciscarl2/gate17.py	H	paper/make_paper_figures.py	B
ciscarl2/gate18.py	H	paper/build_ieee_detailed.py	B
ciscarl2/gate19.py	E	paper/build_model_creation_detailed.py	B
ciscarl2/gate20.py	E	paper/report_content.py	B
ciscarl2/gate21.py	E	paper/make_step_exhibits.py	B
ciscarl2/gate22.py	H	paper/build_code_guide.py	B

ciscarl2/gate23.py	E	paper/code_guide_content.py	S

Part D — Plain-language glossary

- **Estimand:** the exact quantity being estimated. Here it is the difference in 730-day survival probability between treatment and control.
- **AIPW:** a doubly robust way to estimate a treatment effect. It combines a prediction model with the observed outcome, so it can still be right if one model is wrong.
- **Cox model:** a common survival model that estimates how patient characteristics change the risk of the event over time.
- **Censoring:** when a patient's event is not observed before the end of follow-up or before they leave the study.
- **Cross-fitting:** splitting the data into folds so a patient's own outcome is not used to predict that patient. It prevents optimistic bias.
- **CART:** a small decision tree that splits patients into groups using simple yes/no rules.
- **Leaf / rule:** one end point of the tree. Its path from the root is an if-then rule.
- **Support:** the number of discovery patients who satisfy a rule. A small support means the rule may be a lucky accident.
- **Purity:** the share of treated patients in the rule who truly benefit.
- **Capture:** the share of the total true benefit that the rule finds.
- **LCB:** lower confidence bound: the pessimistic end of an interval. The model treats only when it is above zero.
- **Calibration:** adjusting predicted effects so they match observed effects on average.
- **Isotonic calibration:** a flexible calibration that only allows the predicted effect to rise, never fall, as the raw score rises.
- **Influence function:** a formula for the uncertainty of an estimate; it gives the standard error used in the interval.
- **Bootstrap:** repeating an analysis on resampled data to see how much the answer moves.
- **TMLE:** a targeted estimator that updates nuisance predictions toward the effect of interest. The fixed-horizon version was tested and rejected.
- **SCRE:** a survival causal rule ensemble; the external baseline, not the final model.
- **Jaccard:** the overlap between two sets of patients, from 0 (no overlap) to 1 (identical).
- **Operating range:** the sample sizes where the model reliably beats the constant baseline.
- **Abstention:** the model's decision to recommend no treatment when the evidence is too weak.

Where to start

- Read the 31-step report first: it tells the story from the raw file to the frozen model, with the exact output of every step.
- Use this guide when you need to know what a specific file does, or whether a file is part of the final model or only the audit trail.
- Read CANONICAL_RESULTS.md for the numbers and CLAIMS_v1.17.md for what may and may not be said.
- Run python run_all.py --dry-run to see the commands, --quick for a safe smoke test, or the full command to rebuild the evidence.

Every claim in the project can be traced from a code file to a result file to a report sentence. That traceability is the point of this guide.